

# Satyam Awasthi

Researcher & Software Engineer

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## Professional Summary

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Computer Science researcher focused on **Autonomous Systems**, **Spatial Perception**, and **Model-Based and Learning-Based Control**. My research explores how perception, predictive modeling, and planning can be integrated with control frameworks to enable robust long-horizon autonomy in robotic systems operating under uncertainty.

## Research Interests

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Autonomous Systems, Spatial Perception, Model-Based and Learning-Based Control, and Robot Planning, with emphasis on long-horizon decision-making under uncertainty in real-world robotic environments.

## Education

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**University of California, Santa Barbara (UCSB)**

*M.S. in Computer Science (GPA: 3.91/4.00)*

Santa Barbara, CA

Sep 2021 – Jun 2023

**Indian Institute of Technology, Kharagpur (IIT Kharagpur)**

*B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)*

Kharagpur, India

Jul 2016 – Jul 2020

## Research Experience

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**Four Eyes Laboratory, University of California, Santa Barbara**

Sep 2021 – Jun 2023

*Graduate Researcher*

Advisor: [Prof. Tobias Höllerer](#), Co-Advisor: [Prof. Michael Beyeler](#)

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

**Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur**

Jan 2016 – Dec 2018

*Undergraduate Researcher*

- Contributed to the development of **Eklavya 6.0**, the IIT Kharagpur autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**. [Competition Video](#) · [Team AGV](#)
- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A\*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.

## Publications

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### Peer-Reviewed Conference Papers

**SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing**

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

*COMSNETS*, 2022. DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

### Workshop & Poster Publications

**Eye Tracking Performance in Mobile Mixed Reality**

**Satyam Awasthi**, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

*IEEE VRW*, 2024. DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

## EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

ISMAR Adjunct, 2023. DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

## Research Artifacts

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### EyeTTS Calibration Framework

2022–2024

[Calibration Framework](#) · [User Study Framework](#)

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

### EyeTTS Mixed-Reality Eye Tracking Dataset

2022–2024

[Dataset](#) · [User Study Videos](#)

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

## Selected Projects

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### JitterNot

Jan 2022 – Mar 2022

*Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)*

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

### IGVC Lane Detection & Navigation

June 2018

*ROS-Based Lane Perception Pipeline for Autonomous Ground Vehicles (IIT Kharagpur)*

- Developed a ROS-integrated lane perception pipeline using OpenCV, bird's-eye-view projection, CUDA/GPU-accelerated SLIC-style superpixel segmentation, and parabolic lane fitting for real-time autonomous navigation.
- Rejected obstacle and non-lane regions using contour-based ROI extraction, HOG descriptors, and SVM classification, then published local waypoint goals and segmented top-view lane masks for ROS navigation.

### In Motion — HAR using DeepConvLSTM

Nov 2019

*Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)*

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

### Battleship Solitaire Solver

Apr 2022 – Jun 2022

*Solver-Aided Constraint Synthesis for Structured Search Problems (UCSB)*

- Formulated Battleship Solitaire as a symbolic constraint-satisfaction problem in Rosette/Racket, encoding grid cells, ship placements, row/column tallies, adjacency rules, and revealed-cell constraints.
- Used Rosette's solver-aided execution to synthesize valid board configurations satisfying all puzzle constraints.

## Industry Experience

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### Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

*Software Engineer 2*

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

### Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

*Software Dev Engineer II (IC3)*

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

## Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

*Software Engineering Intern*

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

## Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

*Software Engineer (CL 2)*

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

## Teaching

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### University of California, Santa Barbara

Santa Barbara, CA

*Graduate Teaching Assistant*

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

## Technical Skills

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**Programming:** Python, C++, Java, Kotlin, MATLAB

**Robotics & Simulation:** ROS/ROS2, Gazebo, RViz, Unity

**Machine Learning & Vision:** PyTorch, OpenCV, CUDA/GPU Acceleration

**Methods:** State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

**Developer Tools:** Git

## Academic Service & Peer Review

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### Peer Reviewer, UIST 2024

*ACM Symposium on User Interface Software and Technology*

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.